4 THE LOVE OF BUBBLES

Game Development Brief

Unity + Codex / GPT-6 Astra Medium + Blender

Authoritative design and implementation brief for a stylized 3D tycoon / management game based on the real 4 The Love Of Bubbles handmade bath, body, skin, and hair-care business.

Core product statement
The player starts with a small home-based handcrafted body-care business, grows into a successful first retail store, then expands into a string of stores while preserving product quality, brand identity, customer loyalty, and the handmade spirit of the company.

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1. Project Intent and Non-Negotiables

4 The Love Of Bubbles is a warm, stylized 3D tycoon and business-management game created as a personal gift. It should feel like a legitimate management-simulation game rather than a novelty project. The real business identity is central to the experience.

Primary design goals

- Make growth visible. The player should physically see the business evolve from a cramped home setup into a polished retail chain.
- Make products matter. Recipes, ingredients, product quality, packaging, inventory, and customer preferences remain meaningful throughout the game.
- Make management satisfying. The player solves production bottlenecks, stock problems, staffing issues, layout problems, pricing questions, and expansion decisions.
- Keep art attractive but efficient. Visual quality comes from clean composition, appealing lighting, readable silhouettes, charming environments, and polished UI instead of expensive rendering.
- Preserve the identity of 4 The Love Of Bubbles. Expansion should not erase the handcrafted origin of the brand.
- Build incrementally. Do not attempt the complete game in the first Codex task.

Non-negotiable rules

- Engine: Unity.
- Primary coding agent: Codex using GPT-6 Astra at Medium reasoning for normal development tasks.
- 3D content creation: Blender where custom assets are required.
- Rendering pipeline: Unity URP.
- Camera: elevated three-quarter management camera with rotation and zoom.
- Art direction: stylized contemporary 3D, readable and warm, with believable human proportions.
- The player must interact with a simulated business space. The game is not primarily menu-only.
- The first implementation target is the Home Business vertical slice.

2. Player Fantasy and Core Loop

The fantasy is to begin as the hands-on founder of a small handmade body-care company and gradually become the owner of a thriving retail brand. The player's role evolves from maker, to shop owner, to manager, to chain operator.

Core gameplay loop

Step	Player Action	System Response	Reward	Next Pressure
1	Acquire ingredients	Inventory increases; cash decreases	Ability to produce	Ingredient cost and supply
2	Choose product /	Production job	Batch enters	Time and station

	recipe	created	process	capacity
3	Produce and package	Finished goods created	Sellable inventory	Storage and labor
4	Sell / fulfill demand	Cash and reputation change	Revenue and loyalty	Stockouts and expectations
5	Reinvest	Upgrade tools, staff, space, marketing	Higher capacity / margins	New complexity
6	Expand	Unlock storefronts and systems	Visible business growth	Management and logistics

3. Game Progression Structure

Each era introduces new responsibilities rather than simply making the numbers larger.

Era I: Home Business

Hands-on production in a small home workspace. The founder personally creates products, packages orders, purchases ingredients, and manages limited space.

- Basic recipes
- Ingredient inventory
- Workstations
- Order fulfillment
- Online / local demand
- Equipment upgrades
- Storage pressure

Era II: First Retail Store

The first physical 4 The Love Of Bubbles store introduces walk-in customers, store layout, merchandising, employees, checkout, staffing, and retail reputation.

- Retail floor
- Customer pathing
- Shelves / displays
- Checkout
- Staff
- Cleaning
- Store design

- Foot traffic
- Local marketing

Era III: Retail Chain

The player manages multiple locations and increasingly operates at the brand level instead of personally handling every task.

- Multiple stores
- Store managers
- Central production
- Warehouse / distribution
- Store-specific demand
- Wholesale
- Regional strategy
- Brand-wide marketing

4. Home Business Phase

This is the first playable vertical slice. Its purpose is to prove the economy, production, inventory, UI, and save systems before retail AI is added.

Starting environment

- One compact home workshop / kitchen-production room.
- One founder character or founder avatar representation.
- One or two production stations.
- One ingredient storage area.
- One finished-goods storage shelf.
- One packing / shipping station.
- Laptop or order-management interface.
- Simple props that make the room feel lived-in and increasingly crowded.

Initial product set

- Body Butter
- Handmade Soap
- Body Scrub

First 30 minutes

1. Player receives a short introduction and sees the initial cash balance.
2. Player checks available ingredients.
3. Player queues a small batch.
4. Production consumes ingredients over time.
5. Finished units enter product inventory.
6. Customer orders appear.

7. Player fulfills orders and receives revenue.
8. Player purchases additional ingredients.
9. Player buys the first equipment upgrade.
10. Production becomes measurably faster or more efficient.
11. Orders increase and the room becomes increasingly constrained.
12. A milestone indicates that the business is outgrowing the home setup.

5. First Store Phase

The first store converts the game from production management into a full tycoon experience.

New systems

- Store build mode with walls, doors, shelving, counters, signage, plants, lighting, and decorative fixtures.
- Walk-in customers with browsing, selection, queueing, purchasing, satisfaction, and abandonment behavior.
- Shelf inventory and manual / automatic restocking.
- Checkout station and queue management.
- Employee hiring, wages, role assignment, schedules, training, and productivity.
- Store cleanliness and presentation.
- Local marketing and customer traffic.
- Customer reviews and store reputation.
- Merchandising: product placement influences visibility and sales.

Customer archetypes

Archetype	Primary Motivation	Typical Demand
Gift Shopper	Packaging, bundles, fragrance, presentation	Gift sets and premium displays
Natural-Ingredient Shopper	Ingredient quality and botanical positioning	High-quality formulations
Value Shopper	Price and promotions	Affordable staples
Hair-Care Shopper	Specialized hair needs	Hair-care line
Skincare Enthusiast	Premium ingredients and reviews	Higher-end formulas
Loyal Regular	Consistency and availability	Repeat-purchase staples
Trend Shopper	Popularity and marketing	Seasonal / viral products

6. Multi-Store / Chain Phase

- Choose new store locations based on rent, local demand, customer profile, foot traffic, and expected margins.
- Appoint store managers and review store KPIs.
- Transfer inventory between locations.
- Open a centralized production facility once the chain is large enough.
- Use distribution scheduling to deliver stock to stores.
- Forecast demand to prevent one store from running out while another is overstocked.
- Take optional wholesale contracts with salons, boutiques, spas, hotels, or other businesses.
- Develop seasonal launches, premium lines, gift sets, and regional product mixes.
- Maintain a company-wide quality / identity metric.

7. Product, Recipe, and Ingredient Systems

Recipe model

Recipes should feel creative without becoming a chemistry simulator. Each recipe combines a product base, ingredients, optional modifiers, production time, yield, cost, and customer-facing attributes.

- Moisturizing / conditioning value
- Fragrance appeal
- Natural / botanical quality
- Sensitivity friendliness
- Luxury / premium appeal
- Production cost
- Production time
- Shelf life
- Customer rating potential

Ingredient data

- Name
- Category
- Base cost
- Supplier
- Lead time
- Stock quantity
- Quality tier
- Recipe modifiers
- Optional seasonality / scarcity

Product data

- Product ID

- Display name
- Product family
- Recipe ID
- Container / packaging type
- Unit production cost
- Retail price
- Inventory count
- Popularity
- Average rating
- Brand-fit score
- Seasonality
- Unlock status

Recipe book

Successful formulas can be saved permanently. Players should be able to duplicate a formula and create variants instead of rebuilding from scratch.

8. Customers, Staff, Reputation, and Brand Identity

Customer simulation

- Each customer has a budget, patience value, product preferences, price sensitivity, and brand affinity.
- Customers enter with one or more shopping intents.
- Customers evaluate displays and available stock.
- Customers can ask for help if staff are available.
- Customers may purchase multiple items.
- Satisfaction is affected by availability, price, wait time, cleanliness, staff service, and store appeal.
- Satisfied customers contribute to reputation and repeat traffic.
- Unsatisfied customers may leave without buying or generate negative review pressure.

Staff system

- Roles: Sales Associate, Cashier, Stocker, Production Worker, Store Manager, later Regional Manager.
- Stats: Speed, Service, Reliability, Production Skill, Experience.
- Costs: Wage, training cost, scheduled hours.
- Needs should be lightweight. Avoid turning the game into a life simulator.
- Training creates visible productivity improvements.

Brand identity / Brand Soul

Use a persistent company-quality metric, working title Brand Soul. It represents whether the expanding company still feels like the handcrafted, quality-focused business customers originally loved.

- Positive: quality ingredients, consistent formulas, trained staff, original product development, attractive stores, good service.
- Negative: excessive cost-cutting, frequent stockouts, poor quality, dirty stores, weak service, careless overexpansion.
- Growth itself is not negative. The player can build a large company while preserving the brand.

9. Economy and Balancing Model

Revenue	Costs
Direct product sales	Ingredients
Online orders	Packaging
Store sales	Staff wages
Wholesale contracts	Rent / lease
Seasonal launches	Utilities
Gift sets / premium bundles	Marketing
	Equipment maintenance / replacement
	Distribution / logistics

Balancing principles

- Early constraint: cash and production capacity.
- Midgame constraint: space, staffing, service, and stock availability.
- Late-game constraint: logistics, forecasting, store performance, management overhead, and brand consistency.
- Money should eventually become less dominant as operational complexity increases.
- Avoid runaway exponential growth that removes meaningful decisions.
- Show gross revenue, expenses, net profit, margin, and major cost categories clearly.

10. Visual Design Bible

Global art directive
4 The Love Of Bubbles is a warm, stylized contemporary 3D tycoon game. Maintain believable human proportions, clean moderate-poly geometry, soft materials, warm retail lighting, readable silhouettes, strong product color, uncluttered environments, and an elevated rotatable management camera. Favor clarity, charm, and simulation feedback over graphical complexity.

Camera

- Perspective camera, not strict orthographic.
- Downward angle approximately 35 to 45 degrees.
- 360-degree rotation.
- Smooth zoom from detail view to whole-store management view.
- Default play view keeps characters and shelves readable.

Characters

- Believable stylized proportions, approximately 7 to 7.5 heads tall.
- Avoid huge-head caricatures.
- Modular bodies, clothing, hairstyles, skin tones, and accessories.
- Founder receives a higher-detail face, additional outfits, and more unique animation.
- Use a reusable animation library for customers and staff.

Environment

- Modular walls and furniture.
- Natural wood, painted surfaces, glass, ceramic, cardboard, brushed metal, fabric, and greenery.
- Retail architecture stays restrained so colorful products remain the visual focus.
- Home workshop becomes increasingly crowded as demand grows.
- The first store should feel like a major visual reward.

Lighting and materials

- Unity URP.
- Warm, soft retail lighting.
- Soft daylight through windows.
- Moderate ambient occlusion and contact shadows.
- Avoid extreme bloom, fog, aggressive depth of field, or heavy post-processing.
- Most props: 512 or 1024 textures. Hero assets: 2048 where justified.

Product mesh families

- Jars: small, medium, large.
- Bottles: pump, squeeze, dropper, spray.
- Tubs: shallow and standard.
- Soap: bar, wrapped bar, boxed bar.
- Specialty: deodorant, scrub container, hair-care container, bath bomb, gift box.
- Use material, label, cap, and color variation to create a large catalog from a small mesh library.

11. UI / UX and Information Design

UI clarity has higher priority than visual spectacle. The player must understand why the business is succeeding or failing.

Main HUD

- Top: Cash, Daily Profit, Reputation / Brand Soul, Customer Count, Date / Time.
- Left navigation: Build, Products, Production, Inventory, Staff, Marketing, Research / Development, Reports.
- Right inspector: contextual details for selected shelves, workstations, employees, customers, rooms, or stores.
- Contextual build toolbar while build mode is active.

Required overlays

- Customer traffic heat map
- Sales / revenue overlay
- Low-stock / inventory overlay
- Staff coverage overlay
- Customer satisfaction overlay
- Cleanliness overlay
- Profitability overlay

Use brief icons and tooltips for satisfaction, price complaints, waiting, missing stock, cleanliness, and product excitement. Avoid permanent icon clutter.

12. Unity Technical Architecture

Recommended project configuration

- Unity LTS unless a specific feature requires another version.
- Universal Render Pipeline (URP).
- New Input System.
- TextMeshPro.
- AI Navigation / NavMesh for early customer movement unless profiling proves another solution is needed.
- Cinemachine is optional; a custom management camera is acceptable.
- Use ScriptableObjects for static product, ingredient, recipe, furniture, customer-archetype, and staff-role data.
- Use runtime state classes / save DTOs for mutable values. Do not store active save-state directly in ScriptableObject assets.

Core systems / services

System	Responsibility
GameBootstrap	Starts services, loads configuration, creates new game or loads save.
TimeService	Game clock, pause, speed controls, day

	transitions.
EconomyService	Cash, transactions, expenses, revenue ledger.
InventoryService	Ingredient and finished-product quantities.
RecipeService	Recipe definitions, unlocks, variants, batch calculations.
ProductionService	Queues, workstation jobs, ingredient consumption, output creation.
OrderService	Home-era customer orders and fulfillment.
CustomerService	Retail-era spawning, archetypes, needs, satisfaction.
StaffService	Employees, roles, wages, schedules, skill progression.
StoreService	Store configuration, ownership, local stats, store-level KPIs.
BuildService	Place, rotate, move, sell, and validate furniture / stations.
ReputationService	Reviews, reputation, Brand Soul.
SaveService	Versioned save/load, autosave, backup.
UIService	High-level UI state and panels.
Developer Diagnostics	Simulation values, debug commands, test-state tools.

Suggested Unity folder structure

Assets/_Project/Art

Assets/_Project/Audio

Assets/_Project/Data

Assets/_Project/Prefabs

Assets/_Project/Scenes

Assets/_Project/Scripts/Core

Assets/_Project/Scripts/Simulation

Assets/_Project/Scripts/AI

Assets/_Project/Scripts/Building

Assets/_Project/Scripts/UI

Assets/_Project/Scripts/Save

Assets/_Project/Tests

Assets/_Project/ThirdParty

Coding rules

- Prefer clear, testable C# over clever abstractions.
- Do not create a giant GameManager containing unrelated systems.
- Use interfaces only where they improve testability or substitution.
- Keep static data separate from runtime state.
- Avoid Update() polling on large numbers of objects. Use events, timers, pooled update groups, or scheduled ticks where practical.
- Centralize money transactions so the economy can be audited.
- Centralize save serialization and versioning.
- Every major system should expose developer diagnostics.

13. Blender Asset Pipeline

Modeling standard

- 1 Blender unit = 1 meter.
- Apply transforms before export.
- Use clean origins and pivots.
- Maintain consistent orientation and scale.
- Use simple, clean topology.
- Keep UVs only as complex as required.
- Reuse material families.
- Avoid unnecessary hidden geometry.
- Keep modular dimensions compatible with the Unity placement grid.

Asset naming examples

FTLOB_Shelf_Wood_01

FTLOB_Counter_Checkout_01

FTLOB_Display_Table_Round_01

FTLOB_Product_Jar_Medium_01

FTLOB_Product_Bottle_Pump_01

FTLOB_Production_Mixer_01

FTLOB_Prop_ShippingBox_01

FTLOB_Character_Customer_F_01

Approximate geometry budgets

- Small prop: 200 to 1,500 triangles.
- Furniture: 1,000 to 5,000 triangles.
- Production equipment: 3,000 to 10,000 triangles.
- Typical NPC: 10,000 to 20,000 triangles.
- Founder character: 20,000 to 30,000 triangles if useful.
- These are budgets, not mandatory targets.

Asset workflow

13. Block out the asset in Blender.
14. Verify scale against a human reference and 1-meter grid.
15. Finalize geometry and UVs.
16. Create simple material assignments.
17. Export FBX or use a Unity-compatible Blender import workflow.
18. Create a Unity prefab.
19. Set colliders, interaction points, shelf sockets, placement footprint, and optional LODs.
20. Test at gameplay camera distance before polishing.

14. Astra / Codex Operating Instructions

GPT-6 Astra is suited to long multistep software work and can fill routine implementation gaps. This brief remains authoritative so consequential game-design decisions stay under human control.

Recommended use of Astra Medium

- Use Medium for routine Unity implementation, debugging, refactoring, editor tooling, tests, data pipelines, and project maintenance.
- Give Astra one milestone or coherent subsystem at a time.
- Allow Astra to make routine engineering decisions that do not materially change player-facing design.
- Require Astra to flag consequential design conflicts rather than silently redesigning the game.
- Ask Astra to inspect the current project before modifying architecture.
- Require tests or reproducible validation for economy, inventory, save, and production logic.
- Require a concise change log after substantial tasks.
- Prefer finishing a complete vertical slice over creating many half-implemented systems.

Astra may decide without approval

- Exact private method names.
- Small internal helper classes.
- Reasonable serialized-field organization.
- Editor-only debug tooling.

- Minor UI spacing when no exact design has been specified.
- Low-level implementation details that preserve documented behavior.

Astra must not redesign autonomously

- Core progression: home business -> first store -> retail chain.
- The physical 3D management-game presentation.
- The recipe / product-management focus.
- The small first milestone.
- The stylized contemporary visual direction.
- The role of Brand Soul / quality identity.
- Major economy rules, camera philosophy, or store simulation behavior.
- Major dependencies or paid packages without a clear reason.

Codex task protocol

21. Read this brief first.
22. Inspect the Unity project and existing files before making assumptions.
23. State the implementation target briefly.
24. Implement the requested milestone completely.
25. Run available tests and compile checks.
26. Fix compile errors caused by the task.
27. Do not expand scope merely because another system would be useful later.
28. At completion, report files changed, features implemented, tests run, assumptions, and blockers.

15. Milestones and Vertical Slice Plan

Milestone 0: Project Foundation

- Create Unity project with URP and core folders.
- Create bootstrap scene and home-workshop scene.
- Implement game clock, pause, and speed controls.
- Implement basic data architecture using ScriptableObjects.
- Implement economy ledger.
- Implement versioned save/load framework.
- Create developer diagnostics panel.

Milestone 1: Home Business Vertical Slice

- Three products: Body Butter, Handmade Soap, Body Scrub.
- Small ingredient set.
- One or two production stations.
- Production queue.
- Ingredient consumption.
- Finished-goods inventory.
- Simple customer order generation.

- Order fulfillment.
- Revenue and expenses.
- One equipment upgrade.
- Basic HUD.
- Save/load.
- Visible milestone for outgrowing the home.

Milestone 2: Home Business Polish

- Improve room art and product props.
- Add founder movement or interaction presentation.
- Add sound feedback.
- Add tutorial prompts.
- Add recipe book and product variants.
- Add basic marketing and increasing demand.
- Balance the first 30 to 45 minutes.

Milestone 3: First Store Prototype

- Basic store map.
- Customer spawning and navigation.
- Shelving and product displays.
- Customer browsing and purchase logic.
- Checkout queue.
- Basic staff.
- Store reputation.
- Simple build mode.

Milestone 4: First Store Full Loop

- Restocking.
- Cleaning.
- Staff schedules and training.
- Expanded customer archetypes.
- Store design score.
- Marketing.
- Additional products.
- Store profitability reports.

Milestone 5: Expansion / Second Store

- Location selection.
- Second-store ownership.
- Store manager system.
- Separate inventory and KPIs.
- High-level store switching.
- Transfer inventory.

Milestone 6: Chain Management

- Central production facility.
- Distribution.
- Wholesale contracts.
- Regional managers.
- Brand-wide reports.
- Late-game expansion goals and ending milestone.

16. Testing, Save Data, and Performance

Testing priorities

- Economy transactions cannot unintentionally create or destroy money.
- Production consumes the correct ingredients and produces the correct output.
- Inventory never becomes negative.
- Canceled production jobs refund or preserve ingredients according to documented rules.
- Save and load reconstruct the same business state.
- Versioned saves can be migrated when data formats change.
- Retail customers cannot become permanently stuck in invalid states.
- Staff tasks recover if a target object is removed.
- Store profit reports reconcile with the transaction ledger.

Performance target

- Primary target: 1080p at 60 FPS on modest modern hardware.
- Do not design around an RTX-class GPU requirement.
- Initial store target: approximately 30 to 50 active customers.
- Profile before attempting 100+ customers.
- Optimize pathfinding, AI updates, UI refresh rates, and draw calls before obsessing over polygon counts.
- Use object pooling where repeated spawning warrants it.

Save requirements

- Manual save.
- Autosave.
- At least one rolling backup.
- Save version number.
- Graceful handling of missing or obsolete optional data.
- Developer option to create test saves at major progression phases.

17. Scope Control and Explicit Exclusions

Outside initial scope unless deliberately added later:

- Open-world city simulation.
- Driving gameplay.
- Complex life-simulation needs for every NPC.
- Advanced facial animation or dialogue cinematics.
- Multiplayer.
- Online services.
- Real external e-commerce integrations.
- Photorealistic graphics.
- Large procedural cities.
- Complex manufacturing chemistry simulation.
- Hundreds of unique 3D product meshes.
- Console support during the early prototype.
- Mod support during the early prototype.

Scope rule
Whenever a new feature is proposed, ask whether it improves the core loop of making products, satisfying demand, managing a store, and expanding the business. If not, defer it until the vertical slice is proven.

18. Exact Starter Prompt for Astra Medium

Use this after making this document available to Codex:

You are the primary implementation agent for a Unity game titled 4 The Love Of Bubbles.

Read the attached Game Development Brief completely before making changes. Treat it as the authoritative project specification.

The game is a stylized 3D tycoon / management simulation based on a real handmade bath, body, skin, and hair-care business. The campaign progression is:
1. Home business
2. First physical retail store
3. Multiple-store retail chain

Do not attempt to build the entire game now.

Your current target is Milestone 0: Project Foundation, followed by Milestone 1 only when the foundation is stable.

Before editing:
- Inspect the existing Unity project structure, packages, scenes, and scripts.
- Preserve anything already working unless it conflicts with the brief.
- Identify any technical conflict that would materially change the design. Do not silently redesign player-facing systems.

Implementation requirements:

- Unity with URP.
- Clean modular C# architecture.
- ScriptableObjects for static product, ingredient, recipe, customer-archetype, and furniture data.
- Runtime state must be separate from ScriptableObject assets.
- Central economy transaction ledger.
- Versioned save/load architecture.
- Game clock with pause and time-speed controls.
- Developer diagnostics.
- Avoid giant manager classes and unnecessary dependencies.
- Do not use paid packages without explicit approval.
- Favor a complete, testable vertical slice over broad but incomplete systems.

For Milestone 1, the initial playable loop is:
Acquire ingredients -> queue product batch -> consume ingredients -> produce finished goods -> receive customer orders -> fulfill orders -> earn money -> purchase more ingredients -> buy an equipment upgrade -> improve production.

The initial product set is:
- Body Butter
- Handmade Soap
- Body Scrub

The first milestone requires:
- A small home-workshop scene
- Ingredient inventory
- Finished-goods inventory
- Production queue
- One or two production stations
- Order generation and fulfillment
- Revenue and expenses
- Basic HUD
- Save/load
- One equipment upgrade
- A visible progression milestone showing that the business is outgrowing the home workspace

Do not add retail customers, multiple stores, complex build mode, or late-game systems during Milestone 1.

Work incrementally. Compile and test after meaningful implementation batches. Fix errors introduced by your changes.

At completion, report:
1. What you implemented
2. Files created or changed
3. Tests / compile checks run
4. Any design assumptions you made
5. Any blockers or recommended next task

If a routine implementation detail is unspecified, choose a sensible default and document it. If a decision would materially change gameplay or scope, preserve the brief and flag the decision instead of inventing a new design.

19. Development Checklist

☐ Create or open the Unity project.

☐ Confirm URP.

☐ Update Codex / desktop tooling if required for Astra access.

☐ Place this development brief where Codex can read it.

☐ Create version-control repository before major implementation.

☐ Create project folder structure.

☐ Run the Astra starter prompt.

☐ Complete Milestone 0.

☐ Compile and play-test the foundation.

☐ Create Milestone 1 data assets.

☐ Implement the home production loop.

☐ Test economy and inventory.

☐ Test save/load.

☐ Add minimal art only after the loop works.

☐ Play through the first 20 to 30 minutes.

☐ Adjust pacing and upgrade costs.

☐ Do not begin retail AI until the home loop is enjoyable.

☐ Tag or branch the stable vertical slice before Milestone 2.

Appendix A. Naming Standards

C# type examples

ProductDefinition

IngredientDefinition

RecipeDefinition

ProductionJob

InventoryState

StoreState

CustomerArchetypeDefinition

StaffRoleDefinition

EconomyTransaction

GameSaveData

Asset prefixes

- FTLOB_ = shared project prefix
- SO_ = ScriptableObject asset
- PF_ = prefab
- MAT_ = material
- TEX_ = texture
- UI_ = UI asset
- SFX_ = sound effect
- MUS_ = music
- ANIM_ = animation clip
- SCN_ = scene

Appendix B. Suggested Initial Data Tables

Prototype ingredients

Ingredient	Category	Prototype Use	Notes
Shea Butter	Base	Body Butter	High-value core ingredient
Carrier Oil	Base	Body Butter / Scrub	Generic prototype ingredient
Fragrance / Botanical	Modifier	All three	Prototype scent /

Blend			quality choice
Soap Base / Oils	Base	Soap	Abstracted initial soap ingredient
Sugar / Salt Base	Base	Scrub	Prototype scrub base
Jar	Packaging	Body Butter / Scrub	Packaging inventory
Soap Wrap / Box	Packaging	Soap	Packaging inventory

Prototype product definitions

Product	Batch Size	Prototype Time	Base Cost	Suggested Price
Body Butter	6	45 seconds	Tunable	Tunable
Handmade Soap	8	60 seconds	Tunable	Tunable
Body Scrub	6	40 seconds	Tunable	Tunable

Prototype timing is intentionally compressed for playtesting. Real manufacturing time does not dictate game pacing.

Appendix C. Astra Reference Notes

As of September 2026, OpenAI describes GPT-6 Astra as its most capable model for difficult end-to-end work, including coding and complex multistep workflows. It supports Medium reasoning. OpenAI also notes that Astra can fill routine gaps from context while asking focused questions when a decision can materially change the outcome. For this project, use that ability for engineering details while keeping this development brief authoritative for consequential game-design choices.

Official references:

https://developers.openai.com/api/docs/models/gpt-6-astra

https://developers.openai.com/api/docs/guides/latest-model

https://openai.com/index/gpt-6-astra/

https://help.openai.com/en/articles/20001275